CSE 219
Signals and Linear Systems

Formulas, Methods and Worked Patterns
for the Term Final

What this handbook covers

Part I — Anik Sir (Lectures 1–7)
Signals, LTI systems and convolution, Fourier series, Fourier transform.

Part II — Ashraf Sir
DTFT/DFT, FFT, sampling, Laplace and z-transform.

Built from the lecture slides, the 2022–23 and 2023–24 CSE 219 term finals, the earlier EEE 211/301 and CSE 311 papers on the same content, and both practice sets. Every formula is stated once, followed by the *general procedure* to attack that question type and a worked example taken from a past paper.

How to Use This Handbook

Blue box	The formulas to know by heart.
Green box	The step-by-step procedure for a question type.
Red box	Common traps and sign/limit mistakes.
Amber box	Which TF questions this appears in, and what they ask.
Purple box	A fully worked example from a past paper or the practice sets.

Scope notes. Anik Sir's syllabus excludes slides marked $(*)$ and Parseval's theorem; long foundation derivations are for intuition only. Short proofs are given in compact form (see the *Proofs* boxes). Ashraf Sir's Lecture 0 is history and not examinable.

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Part I

Foundations — the Basics You Need First

Chapter 1

Mathematical Basics for Signals and Systems

Every method in this handbook rests on a short list of tools. If a step in a later solution feels like a "trick", it is almost certainly one of the items below.

1.1 Complex numbers and Euler's formula

Key formulas

$$z = a + jb = re^{j\theta}, \qquad r = |z| = \sqrt{a^2+b^2}, \quad \theta = \operatorname{atan2}(b,a), \qquad j^2 = -1,$$

$$e^{j\theta} = \cos\theta + j\sin\theta, \qquad \cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}, \qquad \sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j},$$

$$z^* = a - jb = re^{-j\theta}, \qquad zz^* = |z|^2, \qquad \frac{1}{z} = \frac{z^*}{|z|^2},$$

$$|z_1 z_2| = |z_1||z_2|, \quad \angle(z_1 z_2) = \angle z_1 + \angle z_2, \qquad \operatorname{Re} z = \frac{z+z^*}{2}, \quad \operatorname{Im} z = \frac{z-z^*}{2j}.$$

Special values to memorise

$$e^{j0} = 1,\ e^{j\pi/2} = j,\ e^{j\pi} = -1,\ e^{-j\pi/2} = -j,\ e^{j2\pi k} = 1,\ e^{j\pi k} = (-1)^k,\ j = e^{j\pi/2},\ \tfrac{1}{j} = -j.$$

$$\frac{1-j}{\sqrt{2}} = e^{-j\pi/4}, \qquad \frac{1+j}{\sqrt{2}} = e^{j\pi/4}, \qquad \sqrt{2}\,e^{j\pi/4} = 1+j.$$

Rectangular ↔ polar

1. Magnitude $r = \sqrt{a^2+b^2}$. Angle: look at the quadrant first (a calculator's $\tan^{-1}$ only gives $-90°..90°$). If $a < 0$ add $\pm\pi$.
2. To *multiply or divide* use polar form; to *add or subtract* use rectangular form.
3. A complex number with $\theta = 0$ or π is real; $\pm\pi/2$ is purely imaginary.

Worked example — amplitude and phase of a ratio

$H(j\omega) = \dfrac{1}{2+j\omega}$ at $\omega = 3$: $|H| = \dfrac{1}{\sqrt{13}} = 0.277$, $\angle H = -\tan^{-1}\frac{3}{2} = -56.3°$. (The phase of a ratio is $\angle\text{num} - \angle\text{den}$.)

1.2 Trigonometric identities

Key formulas

$$\sin^2\theta + \cos^2\theta = 1, \qquad \cos^2\theta = \tfrac{1}{2}(1+\cos 2\theta), \qquad \sin^2\theta = \tfrac{1}{2}(1$$

$$\sin 2\theta = 2\sin\theta\cos\theta, \qquad \cos(A\pm B) = \cos A\cos B \mp \sin A\sin B, \qquad \sin(A\pm B) = \sin$$

$$\cos A\cos B = \tfrac{1}{2}[\cos(A-B)+\cos(A+B)], \quad \sin A\sin B = \tfrac{1}{2}[\cos(A-B)-\cos(A+B)], \quad \sin A\cos B = \tfrac{1}{2}[\sin$$

$$\sin\theta = \cos(\theta - 90^\circ), \qquad -\cos\theta = \cos(\theta \pm 180^\circ), \qquad -\sin\theta = \cos(\theta + 90^\circ).$$

Combining sine and cosine of the same frequency

$$A\cos\omega t + B\sin\omega t = R\cos(\omega t - \phi), \qquad R = \sqrt{A^2+B^2}, \quad \phi = \operatorname{atan2}(B, A).$$

Example: $3\cos 2t + 4\sin 2t = 5\cos(2t - 53.13^\circ)$.

Traps

Do not mix radians and degrees: the phase in $\cos(\omega t + \theta)$ must be in the same unit as ωt. Write $30^\circ = \pi/6$ before substituting numbers into a formula that also has π.

1.3 Series and sums

Key formulas

$$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r} \ (r \neq 1), \qquad \sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \ (|r|<1), \qquad \sum_{n=n_0}^{\infty} r^n = \frac{r^{n_0}}{1-r},$$

$$\sum_{n=0}^{\infty} n r^n = \frac{r}{(1-r)^2}, \qquad \sum_{n=1}^{N} n = \frac{N(N+1)}{2}, \qquad \sum_{n=1}^{N} n^2 = \frac{N(N+1)(2N+1)}{6},$$

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N, & k \equiv 0 \pmod N \\ 0, & \text{otherwise} \end{cases}$$ (roots-of-unity sum: the heart of the DFT).

Evaluating a convolution or z-sum with limits

1. Find the range of the summation index where the integrand/sequence is non-zero (draw $x[k]$ and $h[n-k]$ as step functions in k).
2. Pull out factors independent of k; what remains is a geometric sum $\sum r^k$.
3. Apply $\frac{r^{\text{first}} - r^{\text{last}+1}}{1-r}$; check at one small n by hand.

Worked example — limits of a geometric sum

$\sum_{k=-\infty}^{-1} 12^k = \dfrac{12^{-1}}{1-12^{-1}} = \dfrac{1}{11}$ (first term 12^{-1}, ratio $\frac{1}{12}$ going *downwards*). Always ask: "which end of the sum converges?"

1.4 Calculus you actually use

Key formulas

$$\int e^{at}\,\mathrm{d}t = \frac{e^{at}}{a}, \qquad \int te^{at}\,\mathrm{d}t = \frac{e^{at}(at-1)}{a^2}, \qquad \int_0^\infty t^n e^{-at}\,\mathrm{d}t = \frac{n!}{a^{n+1}}\ (a>0),$$

$$\int e^{at}\cos bt\,\mathrm{d}t = \frac{e^{at}(a\cos bt + b\sin bt)}{a^2+b^2}, \qquad \int e^{at}\sin bt\,\mathrm{d}t = \frac{e^{at}(a\sin bt - b\cos bt)}{a^2+b^2},$$

$$\int_{-\infty}^{\infty}\frac{\mathrm{d}x}{a^2+x^2} = \frac{\pi}{a}, \qquad \int_{-\infty}^{\infty}\frac{\sin x}{x}\,\mathrm{d}x = \pi, \qquad \int_{-\infty}^{\infty}\left(\frac{\sin x}{x}\right)^2\mathrm{d}x = \pi,$$

$$\int_a^b u\,\mathrm{d}v = [uv]_a^b - \int_a^b v\,\mathrm{d}u \qquad \text{(integration by parts: choose } u = \text{the polynomial).}$$

Symmetric intervals

$$\int_{-a}^{a} f_{\text{even}}\,\mathrm{d}t = 2\int_0^a f\,\mathrm{d}t, \qquad \int_{-a}^{a} f_{\text{odd}}\,\mathrm{d}t = 0.$$

Even×even = even, odd×odd = even, even×odd = odd. This is why even signals have only cosine/real Fourier coefficients and odd signals only sines/imaginary coefficients.

Limits and removable singularities

1. 0/0: factor and cancel, or use L'Hôpital (differentiate top and bottom). Famous limits: $\frac{\sin x}{x}\to 1$, $\frac{1-e^{-\epsilon T}}{\epsilon}\to T$, $\frac{\sin(x)}{x}$ at $x=0$ is 1, not 0.
2. $\mathrm{sinc}(0)=1$ and $X(s)=\frac{1-e^{-(s+a)T}}{s+a}$ is finite at $s=-a$ (value T).

1.5 Partial fractions

Key formulas

$$\frac{N(s)}{(s-p_1)(s-p_2)} = \frac{A_1}{s-p_1}+\frac{A_2}{s-p_2}, \qquad A_i = \left[(s-p_i)X(s)\right]_{s=p_i} \quad \text{(cover-up rule).}$$

$$\text{Repeated pole } p \text{ (order } m\text{): } \frac{A_1}{s-p}+\cdots+\frac{A_m}{(s-p)^m}, \quad A_{m-r} = \frac{1}{r!}\frac{\mathrm{d}^r}{\mathrm{d}s^r}\left[(s-p)^m X(s)\right]_{s=p}.$$

Procedure

1. If $\deg N \geq \deg D$, divide first (the quotient is a polynomial: $\delta(t)$ terms for Laplace, a constant/delay for z).
2. Factor the denominator; a quadratic with complex roots stays as $\dfrac{Bs+C}{(s+\sigma)^2+\omega^2}$.
3. Find the constants by the cover-up rule; *check* by evaluating both sides at one convenient value (e.g. $s=0$).
4. For the z-transform, expand $X(z)/z$ (or write in z^{-1} with unit constant term) so that each term is $\frac{A}{1-pz^{-1}}$.

Worked example — cover-up

$\dfrac{s+2}{(s+3)(s+4)}$: at $s=-3$, $A=\frac{-1}{1}=-1$; at $s=-4$, $B=\frac{-2}{-1}=2$. Check at $s=0$: $\frac{2}{12}=\frac{-1}{3}+\frac{2}{4}=\frac{1}{6}$, correct.

1.6 Standard signals: definitions and conventions

Key formulas

$$u(t)=\begin{cases}1, & t>0\\ 0, & t<0\end{cases}, \qquad r(t)=tu(t), \qquad \delta(t)=\frac{\mathrm{d}u}{\mathrm{d}t}, \qquad \int \delta(t)\,\mathrm{d}t=1, \qquad \operatorname{sgn} t=2u(t)-1,$$

$$\operatorname{rect}(t)=\begin{cases}1, & |t|<\frac{1}{2}\\ 0, & \text{else}\end{cases}, \qquad \operatorname{tri}(t)=\max(0,\,1-|t|),$$

$$\operatorname{sinc} x=\frac{\sin \pi x}{\pi x}\ \text{(normalised; used with } f \text{ in Hz)} \qquad \text{or} \qquad \frac{\sin x}{x}\ \text{(unnormalised; used with } \omega).$$

Traps

Always check which *sinc* convention a question uses; the two differ by a factor of π inside the argument, and the energy $\int \operatorname{sinc}^2$ is 1 (normalised) or π (unnormalised). Also $u[n]$ equals 1 at $n=0$, while $u(0)$ is undefined/$\frac{1}{2}$ in continuous time.

1.7 The impulse: rules of the game

Key formulas

$$\int x(t)\delta(t-t_0)\,\mathrm{d}t=x(t_0)\quad\text{(sifting)}, \qquad x(t)\delta(t-t_0)=x(t_0)\delta(t-t_0),$$

$$\delta(at)=\frac{1}{|a|}\delta(t), \qquad \delta(-t)=\delta(t),$$

$$x(t)*\delta(t-t_0)=x(t-t_0), \qquad \delta[n]=u[n]-u[n-1], \qquad x[n]=\sum_k x[k]\delta[n-k].$$

Integrals containing δ

1. Find where the argument of δ vanishes (t_0): is t_0 *inside* the integration limits? If not, the integral is 0; if it sits on a limit, the problem usually says how to treat it.
2. Replace everything else by its value at t_0 (and divide by $|a|$ for $\delta(at)$).
3. If δ is multiplied by a function of t, use $x(t)\delta(t-t_0)=x(t_0)\delta(t-t_0)$ before integrating or transforming.

1.8 Orthogonality and the idea behind Fourier analysis

Key formulas

$$\text{inner product: } \langle x,z\rangle = \int_{t_1}^{t_2} x(t)z^*(t)\,\mathrm{d}t, \qquad \text{orthogonal} \iff \langle x,z\rangle = 0,$$

$$\text{projection/coefficient: } c = \frac{\langle x,\phi\rangle}{\langle \phi,\phi\rangle}, \qquad \text{orthonormal} \iff \langle \phi_i,\phi_j\rangle = \delta_{ij},$$

$$\frac{1}{T}\int_T e^{j(k-m)\omega_0 t}\,\mathrm{d}t = \begin{cases} 1, & k=m \\ 0, & k\neq m \end{cases} \quad \Rightarrow \quad a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}\,\mathrm{d}t.$$

Picture: in vector algebra a coordinate is a dot product with a unit axis. Fourier analysis is the same idea with the axes $e^{jk\omega_0 t}$; a signal is a point with coordinates a_k.

1.9 Systems: what the words mean

One-line tests

linear	$T\{ax_1+bx_2\} = aT\{x_1\}+bT\{x_2\}$ (additivity + homogeneity; zero input $\Rightarrow$ zero output)
time-invariant	delaying the input by t_0 delays the output by t_0: $x(t-t_0) \to y(t-t_0)$
causal	output at t depends on inputs at $\tau \le t$; for LTI, $h(t)=0$ for $t<0$
stable (BIBO)	bounded input $\Rightarrow$ bounded output; for LTI, $\int\lvert h\rvert\,\mathrm{d}t < \infty$ ($\sum\lvert h[n]\rvert < \infty$)
memoryless	output at t depends only on $x(t)$; for LTI, $h = K\delta(t)$
invertible	distinct inputs give distinct outputs

LTI systems: three equivalent descriptions

$$y = x * h \quad \longleftrightarrow \quad Y(\omega) = X(\omega)H(\omega) \quad \longleftrightarrow \quad Y(s) = X(s)H(s)\ (Y(z) = X(z)H(z)).$$

Complex exponentials are eigenfunctions: $e^{j\omega t} \to H(j\omega)e^{j\omega t}$. Cascade: $h_1 * h_2$ (H_1H_2); parallel: $h_1 + h_2$; feedback: $\frac{G}{1+GH}$.

1.10 Energy, power, periodicity in one glance

Key formulas

$$E = \int |x|^2\,\mathrm{d}t \ \left(\sum |x[n]|^2\right), \qquad P = \lim \frac{1}{2T}\int_{-T}^{T} |x|^2\,\mathrm{d}t,$$

$$\text{periodic } T_0 \ \Rightarrow\ P = \frac{1}{T_0}\int_{T_0} |x|^2\,\mathrm{d}t \ \ (E=\infty), \qquad \text{sinusoid } A\cos(\omega t+\theta):\ P = \frac{A^2}{2},$$

$$\text{sum of sinusoids of distinct } \omega:\ P = \sum \frac{A_k^2}{2} \ \text{(powers add)}, \qquad \text{DC } c:\ P = c^2.$$

Periodicity test

1. Continuous time: $\cos(\omega_0 t)$ has $T_0 = 2\pi/\omega_0$; a sum is periodic iff all ratios ω_i/ω_j are rational; $T =$ lcm of the periods ($\omega_0 =$ gcd of the frequencies).
2. Discrete time: $e^{j\omega_0 n}$ is periodic iff $\omega_0/2\pi = m/N$ is rational; $N_0 = N/\gcd(m, N)$.

1.11 Logarithms, decibels and units

Key formulas

$$\text{dB} = 20\log_{10}|H| = 10\log_{10}|H|^2, \qquad -3\ \text{dB} \iff |H| = \tfrac{1}{\sqrt{2}} \iff \text{half power}, \qquad \omega = 2\pi f,\ T = \frac{1}{f}.$$

$$e^{-1} = 0.368, \quad e^{-2} = 0.135, \quad \ln 2 = 0.693, \quad \pi = 3.1416, \quad \sqrt{2} = 1.4142, \quad \sqrt{3} = 1.7321.$$

1.12 Which transform for which problem?

signal type	tool	use for
periodic, continuous	Fourier series a_k, $\omega_0 = 2\pi/T$	line spectra, power, response through a filter ($a_k H(jk\omega_0)$)
aperiodic, continuous	Fourier transform $X(\omega)$	bandwidth, modulation, energy, filtering
finite, discrete	DFT / FFT	numerical spectra, fast circular/linear convolution
discrete, any	DTFT, z-transform	frequency response, difference equations, stability
continuous with initial conditions	Laplace transform	circuits, ODEs, feedback and stability

Part II

Anik Sir — Signals, Systems, Fourier Series and Transform

Chapter 2

Signals: Energy, Transformations, Even–Odd, Impulses, Periodicity

2.1 Energy and power

Key formulas

$$E_{[t_1,t_2]} = \int_{t_1}^{t_2} |x(t)|^2 \, \mathrm{d}t, \qquad E_{[n_1,n_2]} = \sum_{n=n_1}^{n_2} |x[n]|^2,$$

$$P_{[t_1,t_2]} = \frac{E}{t_2 - t_1}, \qquad P_{[n_1,n_2]} = \frac{E}{n_2 - n_1 + 1},$$

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x|^2 \, \mathrm{d}t.$$

Energy signal: $0 < E_\infty < \infty$ (then $P_\infty = 0$). **Power signal:** $0 < P_\infty < \infty$ (then $E_\infty = \infty$). **Neither:** both infinite.
Periodic signal: $P = \frac{1}{T_0}\int_{T_0} |x|^2 \, \mathrm{d}t$. Sinusoid $A\cos(\omega_0 t + \varphi)$: $P = A^2/2$. Complex exponential $Ae^{j\omega_0 t}$: $P = |A|^2$.

Method

1. Decide the behaviour as $t \to \pm\infty$: decaying ⇒ try energy; periodic or settling to a level ⇒ power; growing ⇒ neither.
2. Energy: integrate $|x|^2$ over the support, piece by piece.
3. Power of a signal that settles to c_- and c_+: $P_\infty = (c_-^2 + c_+^2)/2$.
4. Sum of sinusoids of different frequencies: powers add.

Worked example — half-sine, $x = \sin t$ on $[0, \pi]$

$E = \int_0^\pi \sin^2 t \, \mathrm{d}t = \pi/2$, $P_{[0,\pi]} = \frac{1}{2}$. Repeated every π: $E_\infty = \infty$, $P = \frac{1}{2}$; repeated every 2π: $P = \frac{1}{4}$. The signal 1 for $|t| \le 1$, $1/|t|$ beyond has $E = 2 + 2\int_1^\infty t^{-2} \, \mathrm{d}t = 4$ and $P = 0$.

2.2 Time shifting, reversal and scaling

Key formulas

$x(t - t_0)$: delay (right) by t_0. $x(t + t_0)$: advance (left). $x(-t)$: mirror. $x(\alpha t)$: compress by $\alpha > 1$.

$$x(t) \xrightarrow{\text{shift by } \beta} x(t+\beta) \xrightarrow{\text{reverse if } \alpha<0} \xrightarrow{\text{scale by } |\alpha|} x(\alpha t + \beta)$$

A feature of x at $t = a$ appears in $x(\alpha t + \beta)$ at $\boxed{t = (a - \beta)/\alpha}$.

Method

1. Write $x(t)$ piecewise with all break points and values.
2. Rewrite the argument as $\alpha t + \beta$ (expand brackets: $-2(t-1) = -2t+2$).
3. Apply shift, then reversal (only if $\alpha < 0$), then scaling; *name every intermediate signal.*
4. Map every break point with $t = (a-\beta)/\alpha$ and keep the same shapes between them; check one interior point.
5. A sum $x(\cdot) + x(\cdot)$: sketch each term separately, then add point by point.

Traps

$x(2t+3) \neq x(2(t+3))$: a shift made *after* scaling acts on t, not on $2t$. Vertical edges stay vertical; they only move.

Worked example — CSE 219 2022–23 Q7: $x(-2t+2)$ and $x(-t/2+1)$

$x(t) = t+1$ $(-2 \le t < -1)$, 1 $(-1 \le t \le 0)$, $1 - t/2$ $(0 < t \le 2)$.
$x(-2t+2)$: break points $a = -2, -1, 0, 2 \mapsto t = 2, \frac{3}{2}, 1, 0$, so $x(-2t+2) = t$ $(0 \le t < 1)$, 1 $(1 \le t \le \frac{3}{2})$, $3 - 2t$ $(\frac{3}{2} < t \le 2)$.
$x(-t/2+1)$: $t = 2(1-a) = 6, 4, 2, -2$, so $x = \frac{1}{2} + \frac{t}{4}$ $(-2 \le t < 2)$, 1 $(2 \le t \le 4)$, $2 - \frac{t}{2}$ $(4 < t \le 6)$.

In the exam

2023–24 Q1(a) (15): $y(t) = x(-2(t-1)) + x\left(\frac{t+2}{2}\right)$, $x = 1 - |t|$: two triangles, centres $t = 1$ (half-width $\frac{1}{2}$) and $t = -2$ (half-width 2), non-overlapping. 2022–23 Q7 (23). EEE: every year.

2.3 Even and odd parts

Key formulas

$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$, $x_o(t) = \frac{1}{2}[x(t) - x(-t)]$, $x = x_e + x_o$. Even×even=even, odd×odd=even, even×odd=odd; $\int_{-a}^{a}$odd$= 0$; $x_o(0) = 0$.

Method

1. Write $x(t)$ and $x(-t)$ piecewise; the pieces of $x(-t)$ live on the mirrored intervals.
2. Cut the axis at every break point of both and compute $\frac{1}{2}(x \pm x(-t))$ on each piece.
3. Check: x_e is mirror symmetric, x_o is point symmetric; their sum returns x.

Worked example — CSE 219 2022–23 Q5(a)

$f = t+1$ $(-2 \le t \le 0)$, $-\frac{t}{2} + 1$ $(0 < t \le 2)$. For $0 < t \le 2$: $f_e = \frac{1}{2}[(1 - \frac{t}{2}) + (1-t)] = 1 - \frac{3t}{4}$, $f_o = \frac{1}{2}[(1 - \frac{t}{2}) - (1-t)] = \frac{t}{4}$. Hence $f_e = 1 - \frac{3}{4}|t|$, $f_o = \frac{t}{4}$ for $|t| \le 2$ (zero elsewhere).

2.4 Step, impulse and ramp

Key formulas

$$\delta[n] = u[n] - u[n-1], \quad u[n] = \sum_{m=-\infty}^{n} \delta[m] = \sum_{k=0}^{\infty} \delta[n-k],$$

$$\delta(t) = \frac{\mathrm{d}u}{\mathrm{d}t}, \quad u(t) = \int_{-\infty}^{t} \delta(\tau)\,\mathrm{d}\tau, \quad r(t) = tu(t),$$

$$\text{Sifting: } \int_L^R x(t)\delta(t-a)\,\mathrm{d}t = \begin{cases} x(a) & L < a < R \\ 0 & \text{otherwise} \end{cases}, \qquad x(t)\delta(t-a) = x(a)\delta(t-a),$$

$$\delta(at+b) = \tfrac{1}{|a|}\delta\left(t + \tfrac{b}{a}\right), \qquad \int x(t)\delta'(t-a)\,\mathrm{d}t = -x'(a), \qquad u(ct) = u(t)\ (c > 0).$$

Piecewise signal → elementary signals, and its derivative

1. Scan left to right. At each break point a add (new slope − old slope) $r(t-a)$ and (jump) $u(t-a)$.
2. Derivative: replace $r \to u$ and $u \to \delta$. Slopes between jumps give rectangles; each jump J gives $J\delta(t-a)$.
3. Second derivative: impulses at the slope changes (and δ' at jumps).

Worked example — EEE 211 2022–23 Q5(c)

$x = 5t$ on $(0,1)$, -6 on $(1,2)$, $5(t-2)$ on $(2,3)$. At $t=1$: slope $5 \to 0$, jump $5 \to -6$ (-11); at $t=2$: jump $+6$, slope $0 \to 5$; at $t=3$: slope $5 \to 0$, jump -5.

$$x = 5r(t) - 5r(t-1) - 11u(t-1) + 6u(t-2) + 5r(t-2) - 5r(t-3) - 5u(t-3),$$

$$x' = 5[u(t) - u(t-1)] + 5[u(t-2) - u(t-3)] - 11\delta(t-1) + 6\delta(t-2) - 5\delta(t-3).$$

Is $p_\epsilon(t)$ a model of $\delta(t)$?

Check three conditions: (i) area $= 1$ for every ϵ; (ii) $p_\epsilon(t) \to 0$ for $t \neq 0$; (iii) peak $\to \infty$. $p = \frac{1}{\epsilon\pi\cosh(t/\epsilon)}$ and $\frac{1}{\pi}\frac{\epsilon}{t^2+\epsilon^2}$ pass; $\frac{1}{\pi}\frac{\sin \epsilon t}{t}$ fails as $\epsilon \to 0$ (it is a model as $\epsilon \to \infty$).

2.5 Periodicity

Key formulas

CT: $A\cos(\omega_0 t + \varphi)$ has $T_0 = 2\pi/|\omega_0|$. A sum is periodic iff the period ratio is rational; $T = \mathrm{lcm}(T_1, T_2)$.

DT: $x[n] = \cos(\omega_0 n + \varphi)$ is periodic iff $\dfrac{2\pi}{\omega_0} = \dfrac{N}{m}$ is rational; fundamental period $N_0 = N$ in lowest terms. The phase φ never matters.

Worked example — DT periodicity

$2\cos\left(\frac{16\pi}{63}n + \frac{\pi}{3}\right)$: $2\pi/\omega_0 = 63/8 \Rightarrow N_0 = 63$. $\cos 2n$: $2\pi/2 = \pi$ irrational, not periodic.

$x[n] = (3-4j)e^{j13\pi n/7}$: $2\pi/\omega_0 = 14/13 \Rightarrow N_0 = 14$ for both real and imaginary parts; the phase $\theta = \tan^{-1}\frac{4}{3}$ is not a multiple of $\pi/7$, so it cannot be an integer time shift.

Traps

Expand squares and products first ($\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$). $\sqrt{e^{j5\pi t}} = e^{j5\pi t/2}$: halve the frequency. $\delta(\sin t)$ is a train of period π.

Chapter 3

LTI Systems, Convolution, Causality and Stability

3.1 System properties

Key formulas

Linear: $ax_1 + bx_2 \mapsto ay_1 + by_2$. Time-invariant: $x(t - t_0) \mapsto y(t - t_0)$. Memoryless: $y(t)$ uses only $x(t)$. Causal: $y(t)$ uses $x(\tau)$, $\tau \le t$. BIBO stable: bounded in $\Rightarrow$ bounded out. Invertible: distinct inputs $\Rightarrow$ distinct outputs.

Method

1. **Linear**: zero in must give zero out; apply $ax_1 + bx_2$ and compare. An added constant, $|x|$, x^2, $\log x$ break it.
2. **Time-invariant**: compare $S\{x(t - t_0)\}$ with $y(t - t_0)$. Variable coefficients ($t\,x$, $t^2 y$), $x(2t)$, $x(-t)$ give time-variant.
3. **Causal / memory**: list which times of x enter $y(t)$ ($x(t + a)$, $x(-t)$, upper limit $> t$ are non-causal).
4. **Stable**: try $x = 1$ or $x = u(t)$; counterexample, or bound $|y| \le MB$.

Worked example — EEE 211 2019–20 Q1: $y = \frac{1}{3}\int x(\tau)[\delta(t-\tau) + \delta(t-\tau+t_0) + \delta(t-\tau-t_0)]\,\mathrm{d}\tau$

Sifting: $y = \frac{1}{3}[x(t) + x(t + t_0) + x(t - t_0)]$, a three-point moving average. Linear, time-invariant, has memory, non-causal ($x(t + t_0)$), BIBO stable ($|y| \le B$).

Worked example — $\dfrac{\mathrm{d}y}{\mathrm{d}t} + t^2 y = (2t + 3)x$

Linear (first power in x, y, zero initial state). With input $x(t - t_0)$ the coefficients t^2, $2t + 3$ stay while $y(t - t_0)$ satisfies $(t - t_0)^2$, $2(t - t_0) + 3$ — not equal, so **time-variant**.

3.2 Impulse response and the convolution sum

Key formulas

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k], \qquad y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,\mathrm{d}\tau, \qquad x * h = h * x.$$

$x * (h_1 + h_2) = x * h_1 + x * h_2$; cascade $h = h_1 * h_2$; parallel $h = h_1 + h_2$; $x(t) * \delta(t - a) = x(t - a)$. Pure delay: $h[n] = \delta[n - n_0]$. Echo: $h = u[n]$ or $\alpha^n u[n]$ (larger α, slower fade).

Why $y = x * h$ (intuition, 4 lines)

$x[n] = \sum_k x[k]\delta[n-k]$. Time invariance: $\delta[n-k] \to h[n-k]$. Homogeneity: $x[k]\delta[n-k] \to x[k]h[n-k]$. Additivity: $y[n] = \sum_k x[k]h[n-k]$. CT: replace $\sum$ by $\int$ using sifting.

Finite sequences

Length $= L + M - 1$; start index = sum of start indices; $\sum y = (\sum x)(\sum h)$; coefficient of z^n in $A(z)B(z)$ is $\sum_k a[k]b[n-k]$. Example: $\{1,2,1,1\} * \{1,1,-1,2\} = \{1,3,2,2,4,1,2\}$.

3.3 Convolution procedures

DT convolution of infinite sequences

1. Fix n; $h[n-k]$ is h reflected and shifted right by n.
2. Write the support of $x[k]$ and of $h[n-k]$ in k; the overlap gives the summation limits (empty $\Rightarrow y[n] = 0$).
3. Split into cases of n where the limits change. Sum with

$$\sum_{k=0}^{n} \alpha^k = \frac{1-\alpha^{n+1}}{1-\alpha}, \qquad \sum_{k=-\infty}^{m} \beta^k = \frac{\beta^m}{1-\beta^{-1}} \ (\beta > 1).$$

4. Two-sided input: split $x = x_L + x_R$, convolve each, add.
5. Check: support = sum of supports, continuity of the closed form at case boundaries.

Pair	Result	Source
$x = \{0.5, 2\}$, $h = \{1,1,1\}$	$\{0.5, 2.5, 2.5, 2\}$	Lec 4 Ex 2.2
$\alpha^n u[n] * u[n]$	$\frac{1-\alpha^{n+1}}{1-\alpha}u[n]$	Ex 2.3
$2^n u[-n] * u[n]$	2^{n+1} $(n<0)$; 2 $(n \geq 0)$	Ex 2.5
$\left[(\frac{1}{2})^n u[n] + 2^n u[-n]\right] * u[n]$	2^{n+1} $(n<0)$; $4 - 2^{-n}$ $(n \geq 0)$	Ex 2.10
$e^{-at}u * u$	$\frac{1}{a}(1-e^{-at})u(t)$	Ex 2.6
$e^t u(-t) * e^{-t}u(t)$	$\frac{1}{2}e^{-\lvert t\rvert}$	sl. 48–52
$e^{-at}u * e^{-bt}u$	$\frac{e^{-bt}-e^{-at}}{a-b}u(t)$	textbook

Worked example — CSE 219 2022–23 Q8(a), 20 marks

$x[n] = 3^n u[-n-1] + (\frac{1}{3})^n u[n]$, $h[n] = (\frac{1}{4})^n u[n+3]$; $h[n-k] \neq 0$ for $k \leq n+3$. Left part: $\sum_{k \leq \min(-1, n+3)} 3^k 4^{k-n} = 4^{-n} \sum 12^k$. Right part: $4^{-n} \sum_{k=0}^{n+3} (\frac{4}{3})^k$.

$$y[n] = \begin{cases} \frac{256}{11} 3^{n+4}, & n \leq -4, \\ 256 \cdot 3^{-(n+3)} - \frac{32}{11} 4^{-n}, & n \geq -3. \end{cases}$$

Check: $y[-3] = 69.82$ (peak), $y[0] = 6.57$.

Graphical CT convolution — the interval table

1. Flip the simpler signal (usually the rectangle); write its edges as functions of t.
2. List the t-intervals where an edge of one signal passes an edge/break of the other.
3. On each interval write the limits and integrate; if $x = \text{rect}(a, b)$ then $y(t) = H(t-a) - H(t-b)$ with H the running integral of h.
4. Check continuity at interval ends, total support, and area $\int y = \int x \int h$.

Worked example — EEE 211 2023–24 Q2(a): $x = 1$ on $(1, 2)$, $h =$ triangle on $(-1, 1)$

interval	$y(t)$
$t < 0$	0
$0 \le t \le 1$	$t^2/2$
$1 \le t \le 2$	$-t^2 + 3t - \frac{3}{2}$
$2 \le t \le 3$	$(3-t)^2/2$
$t > 3$	0

Area $= 1 \cdot 1 = 1$; peak $\frac{3}{4}$ at $t = \frac{3}{2}$.

Worked example — 2023–24 Q1(b): $(h_1 + h_2) * h_3$ in DT

$h_1 = \delta[n] - \delta[n-2]$, $h_2 = u[n-1]$, $h_3 = \delta[n] - \delta[n-1]$. $h_2 * h_3 = \delta[n-1]$, so $h_{eq} = \delta[n] - \delta[n-2] + \delta[n-3]$. For $x = u[n]$: $y = u[n] - u[n-2] + u[n-3] = \{1, 1, 0, 1, 1, 1, \dots\}$.

Worked example — EEE 211 2023–24 Q1(c): five-system interconnection

Branches h_1, $h_2 * h_3 = (e^{-t} - e^{-2t})u$, $h_4 = \delta$: sum $= \delta + e^{-t}u$. Then $h_{eq} = \frac{1}{2}(e^{-t} + e^{-3t})u$; $\int |h_{eq}| = \frac{2}{3} < \infty$ (stable). $y = (\frac{1}{2}te^{-t} + \frac{1}{4}e^{-t} - \frac{1}{4}e^{-3t})u$.

3.4 Causality and stability

Key formulas

Causal $\Leftrightarrow h(t) = 0$ for $t < 0$ ($h[n] = 0$, $n < 0$). Stable $\Leftrightarrow \int |h|\,\mathrm{d}t < \infty$ ($\sum |h[n]| < \infty$). A nonzero periodic h is neither causal nor stable.

Stability, 3 lines

For $|x| \le B$: $|y(t)| \le \int |h(\tau)||x(t-\tau)|\,\mathrm{d}\tau \le B \int |h|$. Conversely if $\int |h| = \infty$ take $x(-\tau) = \operatorname{sgn} h(\tau)$, so $y(0) = \int |h| = \infty$.

Worked example — CSE 219 2022–23 Q8(b) and 2023–24 Q3(b)

$h_1 = e^{2t}u(-2-t)$: nonzero for $t \le -2$, non-causal; $\int_{-\infty}^{-2} e^{2t} = \frac{1}{2}e^{-4}$, stable. $h_2 = te^{-t}u(t)$: causal, $\int = 1$, stable. $e^{-4t}u(3-t)$: non-causal, $\int_{-\infty}^{3} e^{-4t} = \infty$, unstable. $e^{-6|t|}$: non-causal, $\int = \frac{1}{3}$, stable.

Impulse response of a circuit / ODE

For $y' + ay = bx$: $h = be^{-at}u(t)$. Series RC: $h_C = \frac{1}{RC}e^{-t/RC}u$, $h_R = \delta - \frac{1}{RC}e^{-t/RC}u$. LC low-pass: $LC\,y'' + y = x \Rightarrow h = \omega_0 \sin(\omega_0 t)u(t)$, $\omega_0 = 1/\sqrt{LC}$. General: solve the homogeneous equation for $t > 0$; the impulse fixes $h^{(n-1)}(0^+) = 1/a_n$.

Chapter 4

Fourier Series

4.1 Definition and coefficients

Key formulas

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \qquad a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}\,\mathrm{d}t, \qquad \omega_0 = \frac{2\pi}{T}, \qquad a_0 = \text{average of } x.$$

Trigonometric: $x = a_0 + \sum_{n\geq 1}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$, $a_n = \frac{2}{T}\int x\cos$, $b_n = \frac{2}{T}\int x \sin$. Compact: $C_0 + \sum C_n \cos(n\omega_0 t + \theta_n)$, $C_n = 2|c_n|$, $c_n = \frac{1}{2}(a_n - jb_n)$.

Where a_k comes from (idea)

$\frac{1}{T}\int_T e^{jn\omega_0 t}\,\mathrm{d}t = 1$ if $n = 0$, else 0. Multiply x by $e^{-jk\omega_0 t}$ and average: only the k-th term survives.

By inspection (sum of sinusoids)

Write $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$; collect equal k. Include the constant as a_0. Convert sine to cosine with phase before reading amplitude/phase spectra.

Worked example — $x = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos(2\omega_0 t + \frac{\pi}{4})$

$a_0 = 1$, $a_{\pm 1} = 1 \mp \frac{j}{2}$, $a_{\pm 2} = \frac{1}{2}e^{\pm j\pi/4}$. $|a_1| = \frac{\sqrt{5}}{2}$, $\angle a_1 = -\tan^{-1}\frac{1}{2}$.

Standard waveforms

Rectangular pulse train (1 on $|t| < T_1$, period T): $a_0 = \dfrac{2T_1}{T}$, $a_k = \dfrac{\sin(k\omega_0 T_1)}{k\pi}$. With $T_1 = T/4$: $a_{\pm 1} = \frac{1}{\pi}$, $a_{\pm 2} = 0$, $a_{\pm 3} = -\frac{1}{3\pi}$.

Ramp $x = t$ on $[0, 1)$, $T = 1$: $a_0 = \frac{1}{2}$, $a_k = \dfrac{j}{2\pi k}$.

Triangle ($0 \to 1$ on $[0, 1]$, $1 \to 0$ over $[1, 3]$, $T = 3$): $a_0 = \frac{1}{2}$, $a_k = \dfrac{e^{-jk\omega_0} - 1}{2k^2\omega_0^2}$ ($\omega_0 = \frac{2\pi}{3}$).

Impulse train $\sum \delta(t - mT)$: $a_k = \frac{1}{T}$ for all k (flat); trig form $\frac{1}{T} + \frac{2}{T}\sum \cos\frac{2\pi nt}{T}$.

Traps

The slide prints $6k^2\omega_0^2$ for the triangle; the correct factor is $2k^2\omega_0^2$. a_0 is found separately (area/period).

4.2 Properties

Key formulas

Linearity	$\alpha x + \beta y$	$\alpha a_k + \beta b_k$
Time shift	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0}$
Reversal	$x(-t)$	a_{-k}
Scaling	$x(\alpha t)$	a_k, but $\omega_0 \to \alpha\omega_0$
Differentiation	$\dfrac{\mathrm{d}x}{\mathrm{d}t}$	$jk\omega_0 a_k$
Integration	$\int x\,\mathrm{d}t$	$\dfrac{a_k}{jk\omega_0}$ ($k \neq 0$; find a_0 separately)
Product	$x(t)y(t)$	$c_k = \sum_p a_p b_{k-p}$
Real x		$a_{-k} = a_k^*$
Even x / odd x		a_k real (half-cycle: $\frac{2}{T}\int_0^{T/2} x\cos$) / imaginary ($-\frac{2j}{T}\int_0^{T/2} x\sin$, $a_0 = 0$)
Half-wave symmetry $x(t+\frac{T}{2}) = -x(t)$		$a_k = 0$ for even k

Coefficients of a combined signal

Shift first, then reverse, then scale; update ω_0 after scaling. Example: $x(4-3t) \leftrightarrow a_{-k}e^{-j4k\omega_0}$ with $\omega_0' = 3\omega_0$; $\frac{\mathrm{d}}{\mathrm{d}t}x(4-3t) \leftrightarrow j(3k\omega_0)a_{-k}e^{-j4k\omega_0}$.

Piecewise-linear / piecewise-constant waveform

Differentiate until only impulses remain; an impulse $A\delta(t - t_0)$ in one period contributes $\frac{A}{T}e^{-jk\omega_0 t_0}$. Then $a_k = \dfrac{b_k}{(jk\omega_0)^m}$; get a_0 from the average.

Worked example — CSE 219 2023–24 Q4: odd square wave $\pm A$, $y(t) = x(t - T_0/6)$

$X_k = \dfrac{A}{jk\pi}[1 - (-1)^k] = \dfrac{2A}{jk\pi}$ (odd k), 0 (even k). Half-wave symmetry survives a shift. $Y_k = X_k e^{-jk\omega_0 T_0/6} = \dfrac{2A}{jk\pi}e^{-jk\pi/3}$. $|Y_k| = \dfrac{2A}{\pi|k|}$; $\angle Y_k = -\dfrac{\pi}{2} - \dfrac{k\pi}{3}$ ($k > 0$): $-\frac{5\pi}{6}$, $\frac{\pi}{2}$, $-\frac{\pi}{6}$ for $k = 1, 3, 5$. Trig series: same amplitudes $\frac{4A}{\pi k}$, each harmonic delayed by $k\pi/3$.

Worked example — $p(t) = x^2(t)$, $x = 1 + \cos\omega_0 t$

$a = \{\frac{1}{2}, 1, \frac{1}{2}\}$, $a * a = \{\frac{1}{4}, 1, \frac{3}{2}, 1, \frac{1}{4}\}$ for $k = -2, \ldots, 2$.

4.3 Periodic input to an LTI system

Key formulas

$e^{jk\omega_0 t} \to H(jk\omega_0)e^{jk\omega_0 t}$, so $y_k = H(jk\omega_0)\,a_k$ and $A\cos(\omega t + \theta) \to A|H|\cos(\omega t + \theta + \angle H)$. Average power: $P = \sum_k |a_k|^2$ (apply to the *output* coefficients).

Method

1. Find a_k of the input.
2. Find $H(j\omega)$ (circuit: divider with $j\omega L$, $1/j\omega C$; ODE: $\frac{\mathrm{d}}{\mathrm{d}t} \to j\omega$).
3. Multiply harmonic by harmonic; list magnitude and phase.
4. Ideal LPF with cutoff ω_c keeps harmonics with $|k\omega_0| < \omega_c$.

Worked example — EEE 211 2020–21 Q7(b)

$y'' + 3y' + 2y = x' + x$ gives $H = \dfrac{1}{j\omega + 2}$. For $x = 2\cos t + 6\cos 3t$: $y = \frac{2}{\sqrt{5}}\cos(t - 26.57°) + \frac{6}{\sqrt{13}}\cos(3t - 56.31°)$, average power $\frac{1}{2}\left(\frac{4}{5} + \frac{36}{13}\right) = 1.785$.

In the exam

2023–24 Q4 is a full 30-mark FS problem; EEE 211 asks FS of standard waveforms and ‘periodic input through a circuit’ every year.

Chapter 5

Fourier Transform

5.1 Definition and the key pairs

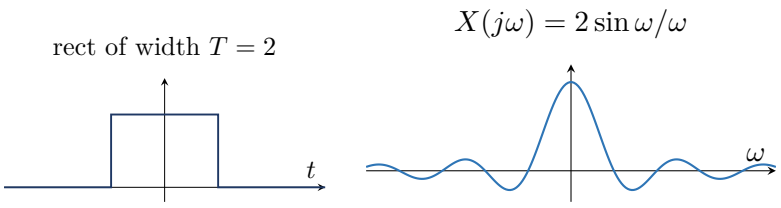
Key formulas

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\,\mathrm{d}t, \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\,\mathrm{d}\omega.$$

From the FS: $Ta_k = X(jk\omega_0)$ (one period of x), periodic x: $X = 2\pi \sum a_k \delta(\omega - k\omega_0)$.

Pair table

$\delta(t)$	1	
1	$2\pi\delta(\omega)$	
$e^{-at}u(t),\ a>0$	$\dfrac{1}{a+j\omega}$	
$e^{-a\lvert t\rvert}$	$\dfrac{2a}{a^2+\omega^2}$	
$te^{-at}u(t)$	$\dfrac{1}{(a+j\omega)^2}$	
$\operatorname{sgn}(t)$	$\dfrac{2}{j\omega}$	
$u(t)$	$\pi\delta(\omega) + \dfrac{1}{j\omega}$	
rect: 1 for $\lvert t\rvert < \frac{T}{2}$	$\dfrac{2\sin(\omega T/2)}{\omega}$	value T at 0
$\dfrac{\sin Wt}{\pi t}$	1 for $\lvert\omega\rvert < W$	
triangle $1 - \frac{\lvert t\rvert}{\tau}$, $\lvert t\rvert < \tau$	$\tau\left[\dfrac{\sin(\omega\tau/2)}{\omega\tau/2}\right]^2$	
$\cos\omega_0 t$	$\pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$	
$\sin\omega_0 t$	$\frac{\pi}{j}[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)]$	
$\sum \delta(t - nT)$	$\dfrac{2\pi}{T}\sum \delta\left(\omega - \dfrac{2\pi k}{T}\right)$	



5.2 Properties

Key formulas

Time shift	$x(t-t_0)$	$e^{-j\omega t_0}X$ (linear phase)
Frequency shift	$x(t)e^{j\omega_0 t}$	$X(j(\omega-\omega_0))$
Modulation	$x(t)\cos\omega_0 t$	$\frac{1}{2}X(j(\omega-\omega_0))+\frac{1}{2}X(j(\omega+\omega_0))$
Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X(j\omega/a)$
Reversal	$x(-t)$	$X(-j\omega)$
Duality	$X(jt)$	$2\pi x(-\omega)$
Differentiation	$\dfrac{\mathrm{d}x}{\mathrm{d}t}$	$j\omega X$
Frequency differentiation	$-jt\,x(t)$	$\dfrac{\mathrm{d}X}{\mathrm{d}\omega}$
Integration	$\int_{-\infty}^{t}x$	$\dfrac{X}{j\omega}+\pi X(0)\delta(\omega)$
Convolution	$x*h$	XH
Multiplication	xy	$\frac{1}{2\pi}X*Y$
Real x		$\lvert X\rvert$ even, $\angle X$ odd; even $x\Rightarrow X$ real

Short proofs

Shift: $\int x(t-t_0)e^{-j\omega t}\,\mathrm{d}t$, $\tau=t-t_0$ gives $e^{-j\omega t_0}X$. *Frequency shift:* $\int xe^{j\omega_0 t}e^{-j\omega t}\,\mathrm{d}t=X(j(\omega-\omega_0))$. *Convolution:* swap integrals, substitute $s=t-\tau$. *Duality:* $2\pi x(-t)=\int Xe^{-j\omega t}\,\mathrm{d}\omega$; rename $t\leftrightarrow\omega$.

Finding the FT of a given signal

1. Match the signal (or pieces of it) to a table pair.
2. Reduce the rest with shift, scaling, modulation, differentiation or reversal; *name each property.*
3. Decompose composite shapes into rectangles/triangles centred at their own centres and multiply each by $e^{-j\omega t_c}$.
4. Sketch spectra: modulation gives two half-height copies at $\pm\omega_0$.

Worked example — CSE 219 2022–23 Q6: $f=e^{-at}u(t)$, $\operatorname{sgn}t$

$f(-t)=e^{at}u(-t)$. $F(j\omega)=\dfrac{1}{a+j\omega}$, $\mathcal{F}\{f(-t)\}=F(-j\omega)=\dfrac{1}{a-j\omega}$. $\operatorname{sgn}t=\lim_{a\to0}[f(t)-f(-t)]$, so by linearity and reversal

$$\mathcal{F}\{\operatorname{sgn}t\}=\lim_{a\to0}\left(\frac{1}{a+j\omega}-\frac{1}{a-j\omega}\right)=\frac{2}{j\omega}.$$

Worked example — CSE 219 2023–24 Q1(c): $\operatorname{rect}\left(\frac{t-\tau}{T}\right)\cos2\pi f_0 t$

$\operatorname{rect}(t)\leftrightarrow\operatorname{sinc}f=\frac{\sin\pi f}{\pi f}$; scaling: $\operatorname{rect}(t/T)\leftrightarrow T\operatorname{sinc}(fT)$; shift: $\times e^{-j2\pi f\tau}$; modulation:

$$X_2(f)=\tfrac{T}{2}\left[\operatorname{sinc}((f-f_0)T)e^{-j2\pi(f-f_0)\tau}+\operatorname{sinc}((f+f_0)T)e^{-j2\pi(f+f_0)\tau}\right].$$

Worked example — 2023–24 Q2: $x_1 = e^{-at}u$, $x_2 = e^{-a|t|}$ and $I(a) = \int \frac{f^2\,\mathrm{d}f}{(a^2+(2\pi f)^2)^2}$

$X_1 = \frac{1}{a+j\omega}$; $x_2 = x_1(t) + x_1(-t) \Rightarrow X_2 = \frac{2a}{a^2+\omega^2}$. (The Parseval part $I(a) = \frac{1}{16\pi^2 a}$ is outside Part-1 syllabus.)

Worked example — CSE 219 2022–23 Q5(b): inverse FT chain

$X = 1$ on $|\omega| < 3\pi$ gives $x(t) = \dfrac{\sin 3\pi t}{\pi t}$. $X_1 = 3j\omega X$ (magnitude $3|\omega|$, phase $\pm\frac{\pi}{2}$) so $x_1 = 3\dfrac{\mathrm{d}x}{\mathrm{d}t} = \dfrac{3(3\pi t\cos 3\pi t - \sin 3\pi t)}{\pi t^2}$.

In the exam

Spectrum sketches with modulation appear in every EEE paper (quadrature multiplexing 2023–24 Q5(a), AM with phase error: output $x(t)\cos\theta$). CSE 311: time delay $\Rightarrow$ linear phase, sampling spectra.

5.3 Filters, distortionless systems, bandwidth

Key formulas

$Y = HX$. Distortionless: $y = Kx(t-t_d) \Leftrightarrow H = Ke^{-j\omega t_d}$ ($|H|$ constant, phase linear through the origin). All-pass: $|H|$ constant but phase need not be linear. Ideal LPF with delay: $h = \dfrac{\sin\omega_c(t-t_d)}{\pi(t-t_d)}$ (non-causal, unrealizable). Compression in time $\Leftrightarrow$ expansion in frequency; derivative widens the spectrum. 3-dB bandwidth of $e^{-at}u$: a rad/s.

Application sentences

Convolution: reverb (convolve with the hall's impulse response), echo $h = \delta[n] + a\delta[n-D]$, smoothing (moving average). **Fourier series**: harmonic analysis, response to periodic sources. **Fourier transform**: denoising, JPEG (drop small high-frequency coefficients), MRI (data in frequency domain), music fingerprinting, vibration diagnosis, ECG notch.

Part III

Ashraf Sir — DFT, FFT and Sampling

Chapter 6

DTFT, DFT and Circular Convolution

6.1 The four Fourier transforms and the DFT pair

	continuous time	discrete time
periodic	CTFS: discrete, aperiodic a_k	DFS/DFT: discrete, periodic $X[k]$
aperiodic	CTFT: continuous, aperiodic $X(j\omega)$	DTFT: continuous, 2π-periodic $X(e^{j\omega})$

Rule: *discrete in one domain* $\Leftrightarrow$ *periodic in the other.*

Key formulas

$$W_N = e^{-j2\pi/N}, \quad W_N^N = 1, \quad W_N^{k+N/2} = -W_N^k,$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn},$$

$$X(e^{j\omega}) = \sum_n x[n] e^{-j\omega n} \ (2\pi\text{-periodic}), \qquad X[k] = X(e^{j\omega})\big|_{\omega=2\pi k/N},$$

$$X[k] = N a_k \ (\text{DFS}), \qquad c_k = X[k]/T_0 \ (\text{CTFS of the sampled signal}).$$

Bins: $k = 0..N/2$ positive frequencies (low→high), $k = N/2..N-1$ negative (high→low). $f_k = kf_s/N$ $(k \le N/2)$, $(k-N)f_s/N$ above; resolution $\Delta f = f_s/N = 1/T_0$.
Zero-padding to $M \ge N$: the same DTFT sampled every $2\pi/M$ (finer picture, no new information).

W_4^{kn}	$n=0$	1	2	3	
$k=0$	1	1	1	1	$W_4 = -j$
$k=1$	1	$-j$	-1	j	
$k=2$	1	-1	1	-1	
$k=3$	1	j	-1	$-j$	

W_8^m	$m=0$	1	2	3
value	1	$\frac{1-j}{\sqrt{2}}$	$-j$	$-\frac{1+j}{\sqrt{2}}$

Worked example — $x = \{1, 2, 3, 4\}$

$X = \{10,\ -2{+}2j,\ -2,\ -2{-}2j\}$; check $X[0] = \sum x$, $X[3] = X^*[1]$; $\sum |x|^2 = 30 = \frac{1}{4}(100{+}8{+}4{+}8)$.

6.2 DFT properties

Key formulas

Circular shift	$x[((n-m))_N]$	$W_N^{km}X[k]$
Modulation	$W_N^{-k_0 n}x[n]$	$X[((k-k_0))_N]$; $(-1)^n x \leftrightarrow X[k-N/2]$
Reversal / conjugation	$x[((-n))_N]$, $x^*[n]$	$X[((-k))_N]$, $X^*[((-k))_N]$
Real x		$X[N-k]=X^*[k]$
Circular convolution	$x \circledast h$	$X[k]H[k]$
Product	$x[n]h[n]$	$\frac{1}{N}X \circledast H$
Parseval	$\sum\lvert x[n]\rvert^2$	$\frac{1}{N}\sum\lvert X[k]\rvert^2$

6.3 Circular convolution and filtering with the DFT

Key formulas

$(x \circledast h)[n] = \sum_{m=0}^{N-1} x[m]\, h[((n-m))_N] \iff X[k]H[k]$. Linear convolution has length $L+M-1$; **circular = linear iff** $N \geq L+M-1$ (zero-pad both).

Linear convolution through the DFT

1. Choose $N \geq L+M-1$ (power of 2 if an FFT is asked); zero-pad x, h.
2. Compute $X[k], H[k]$ (use the W_N table).
3. $Y[k] = X[k]H[k]$ bin by bin.
4. $y[n] = \frac{1}{N}\sum Y[k]W_N^{-kn}$; check against the direct convolution.

Worked example — CSE 219 2023–24 Q5(b): $\{1,2\} * \{1,1\}$ with $N=4$

$X = \{3,\ 1-2j,\ -1,\ 1+2j\}$, $H = \{2,\ 1-j,\ 0,\ 1+j\}$, $Y = \{6,\ -1-3j,\ 0,\ -1+3j\}$. $y[n] = \frac{1}{4}\sum Y[k]j^{kn}$ gives $y = \{1,3,2,0\}$.

Worked example — wrap-around

$x = \{2,4,5,6,5,3,2,1\}$, $h = \{1,2,1\}$: linear $\{2,8,15,20,22,19,13,8,4,1\}$; circular $(N=8)$ $\{6,9,15,20,22,19,13,8\}$ — the tail $\{4,1\}$ folds onto the first two samples.

Chapter 7

Fast Fourier Transform

7.1 Radix-2 decimation in time (DIT)

Key formulas

$X[k] = G[k] + W_N^k H[k]$, $X[k+\frac{N}{2}] = G[k] - W_N^k H[k]$ for $0 \le k < \frac{N}{2}$, where G, H are the $N/2$-point DFTs of the even and odd samples. Inputs in **bit-reversed** order, outputs natural. Cost: direct N^2 multiplications vs FFT $\frac{N}{2}\log_2 N$ (additions $N\log_2 N$); $N = 8$: 64 vs 12; $N = 1024$: 1.05×10^6 vs 5120.
Bit reversal, $N = 8$: $x_0, x_4, x_2, x_6, x_1, x_5, x_3, x_7$.

Radix-2 DIT, every butterfly

1. Zero-pad to $N = 2^m$; write the bit-reversed input.
2. Stage 1: $(a, b) \mapsto (a + b,\ a - b)$ on adjacent pairs.
3. Stage 2: combine 2-point results with $W_4^0 = 1$, $W_4^1 = -j$.
4. Stage 3: combine with $W_8^0 \dots W_8^3$: $X[k] = G[k] + W_8^k H[k]$, $X[k+4] = G[k] - W_8^k H[k]$.
5. Checks: $X[0] = \sum x$, $X[4] = \sum(-1)^n x[n]$, $X[8-k] = X^*[k]$.

Worked example — CSE 219 2023–24 Q5(a): $x = \{1, 2, 3, 4, 3, 2\}$, 8-point

Pad: $\{1, 2, 3, 4, 3, 2, 0, 0\}$; bit-reversed: $1, 3, 3, 0, 2, 2, 4, 0$.
Stage 1: $4, -2 \mid 3, 3 \mid 4, 0 \mid 4, 4$.
Stage 2: $G = (7,\ -2 - 3j,\ 1,\ -2 + 3j)$, $H = (8,\ -4j,\ 0,\ 4j)$.
Stage 3: $W_8^k H[k] = (8,\ -2.828 - 2.828j,\ 0,\ 2.828 - 2.828j)$.

$$X = \{15,\ -4.828 - 5.828j,\ 1,\ 0.828 + 0.172j,\ -1,\ 0.828 - 0.172j,\ 1,\ -4.828 + 5.828j\}.$$

7.2 Decimation in frequency (DIF)

Key formulas

$X[2m] = \mathrm{DFT}_{N/2}\{x[n] + x[n+\frac{N}{2}]\}$, $X[2m+1] = \mathrm{DFT}_{N/2}\{(x[n] - x[n+\frac{N}{2}])W_N^n\}$.
Butterfly: *add/subtract first, then twiddle.* Inputs natural, outputs bit-reversed. Same cost as DIT.

7.3 Matrix (Cooley–Tukey / Bailey) FFT

Key formulas

$N = N_1 N_2$, $n = N_1 n_2 + n_1$ (column-major), $k = N_2 k_1 + k_2$ (row-major):

$$X[N_2 k_1 + k_2] = \sum_{n_1=0}^{N_1-1} W_N^{n_1 k_2} \Big[\sum_{n_2=0}^{N_2-1} x[N_1 n_2 + n_1] W_{N_2}^{n_2 k_2} \Big] W_{N_1}^{n_1 k_1}.$$

Four-step FFT

1. Arrange x column-major in an $N_1 \times N_2$ matrix (radix-2: row 0 = evens, row 1 = odds).
2. N_2-point DFT of every row $\to G$.
3. Multiply entry (n_1, k_2) by $W_N^{n_1 k_2} \to H$.
4. N_1-point DFT of every column; read the result *row-major*.

Worked example — CSE 219 2023–24 Q7(a): $x = \{1, 2, 3, 4, 9, 0, 7, 8\}$

Rows: $\{1, 3, 9, 7\}$, $\{2, 4, 0, 8\}$. $G = \begin{bmatrix} 20 & -8+4j & 0 & -8-4j \\ 14 & 2+4j & -10 & 2-4j \end{bmatrix}$. Twiddles of row 1: $1,\ 0.7071-0.7071j,\ -j,\ -0.7071-0.7071j$, so row 1 of H is $14,\ 4.2426+1.4142j,\ 10j,\ -4.2426+1.4142j$. Column sums and differences give

$X = \{34,\ -3.757+5.414j,\ 10j,\ -12.243-2.586j,\ 6,\ -12.243+2.586j,\ -10j,\ -3.757-5.414j\}$.

7.4 Fast convolution and Bluestein

Key formulas

Fast convolution: pad to $N \geq L + M - 1$, FFT both, multiply, IFFT: $O(N \log N)$ vs $O(LM)$. Bluestein (any N, even prime): $kn = \frac{1}{2}[n^2 + k^2 - (k-n)^2]$ gives

$$X[k] = W_{2N}^{k^2} (a * b)[k], \quad a[n] = x[n] W_{2N}^{n^2}, \quad b[n] = W_{2N}^{-n^2},\ n = -(N-1)..(N-1),$$

$W_{2N} = e^{-j\pi/N}$. Pad to $M \geq 2N - 1$ (power of 2), with the negative-index chirp wrapped to the end so the two ends do not overlap.

Worked example — 3-point DFT of $\{2, -1, 3\}$ by Bluestein

$a = \{2,\ -0.5 + 0.866j,\ -1.5 + 2.598j\}$, $M = 8$; result $X = \{4,\ 1 + 3.464j,\ 1 - 3.464j\}$, equal to the direct DFT.

In the exam

2023–24: Q5(a) 15 marks (8-point DIT), Q7(a) 15 (matrix FFT), Q7(b) 10 (Bluestein derivation); 2022–23 Q4(c) 15 (DFT vs FFT complexity: N^2 vs $\frac{N}{2} \log_2 N$, speed-up $2N / \log_2 N$).

Chapter 8

Sampling and Reconstruction

8.1 Spectrum of the sampled signal

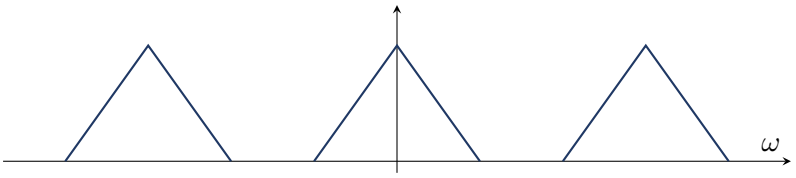
Key formulas

$x_p(t) = x(t)p(t) = \sum x(nT)\delta(t-nT)$, $\omega_s = 2\pi/T$. $P(j\omega) = \dfrac{2\pi}{T}\sum_k \delta(\omega - k\omega_s)$ (impulse train $\to$ impulse train), and

$$\boxed{X_p(j\omega) = \frac{1}{2\pi}X * P = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))}$$

Duality: $X(jt) \leftrightarrow 2\pi x(-\omega)$; product $\leftrightarrow \frac{1}{2\pi}$ convolution. $\delta(\omega - k\omega_s) * X = X(j(\omega - k\omega_s))$.

$X_p(j\omega)$ for $\omega_s > 2\omega_M$: separated copies of height $1/T$



8.2 Sampling theorem and aliasing

Key formulas

Band-limited $X(j\omega) = 0$ for $|\omega| > \omega_M$ is uniquely determined by $x(nT)$ iff $\omega_s > 2\omega_M$ (Nyquist rate $2\omega_M$; $T < \pi/\omega_M$). Product xy: bandwidths add. Aliasing: $f \to |f - kf_s|$.
DT decimation: $X(e^{j\omega}) = 0$ for $W \le |\omega| \le \pi$ allows keeping every M-th sample iff $M \le \pi/W$ (e.g. $W = 2\pi/9 \Rightarrow M = 4$).

Sampling theorem

X_p is X/T repeated every ω_s; copies of width $2\omega_M$ stay apart iff $\omega_M < \omega_s - \omega_M$, i.e. $\omega_s > 2\omega_M$. Then an ideal LPF (gain T, $\omega_M < \omega_c < \omega_s - \omega_M$) isolates X. If $\omega_s < 2\omega_M$ the copies overlap and add; the sum cannot be un-mixed, so x is not recoverable.

Spectrum-sketch questions

1. Sketch $X(j\omega)$; mark ω_M.
2. Compute $\omega_s = 2\pi/T$ and compare with $2\omega_M$.
3. Draw copies at $\pm k\omega_s$ with height $\times\frac{1}{T}$ (natural sampling: copies weighted by $c_k = \frac{\tau}{T}\operatorname{sinc}(\frac{k\tau}{T})$).
4. Overlap $\Rightarrow$ aliasing; cure: anti-aliasing LPF before sampling, higher rate, guard band.

Worked example — EEE 301 2015-16 Q6(b): $z = xy$, 150 Hz and 350 Hz

z is band-limited to 500 Hz, so $T_{s,\max} = 1/(2 \cdot 500) = 1$ ms. $T_s = 0.5$ ms: copies every 2 kHz, no overlap; $T_s = 2$ ms ($f_s = 500$ Hz): copies of width 1 kHz overlap.

Worked example — EEE 211 2019–20 Q7(b)

$x = \frac{\sin(\omega_B t/2)}{\pi t}$ ($X = 1$ on $|\omega| < \omega_B/2$), $p = \sum \delta(t - n\pi/\omega_B)$ ($\omega_s = 2\omega_B$). $Z = \frac{\omega_B}{\pi} \sum X(\omega - 2k\omega_B)$; $h = \frac{\sin(3\omega_B t/2)}{\pi t}$ passes $|\omega| < 1.5\omega_B$, i.e. only the centre copy: $y = \frac{\omega_B}{\pi} x(t)$.

8.3 Reconstruction: sinc, zero-order hold, linear interpolation

Key formulas

	$h(t)$	$H(j\omega)$	properties
Ideal	$\operatorname{sinc}(t/T) = \frac{\sin(\pi t/T)}{\pi t/T}$	T for $\lvert\omega\rvert < \omega_s/2$	non-causal, infinite
ZOH	rect on $[0, T]$	$Te^{-j\omega T/2} \operatorname{sinc}(\frac{\omega T}{2\pi})$	causal; droop, leakage
Linear	triangle on $[-T, T]$	$T \operatorname{sinc}^2(\frac{\omega T}{2\pi})$	non-causal; less leakage

Ideal reconstruction: $x_r(t) = \sum x(nT) \operatorname{sinc}(\frac{t - nT}{T})$. Triangle $= \frac{1}{T} h_0(t + \frac{T}{2}) * h_0(t + \frac{T}{2})$. At $\omega_s/4$ ($T = 1$): $|H_0| = 0.90$, $|H_1| = 0.81$.

In the exam

2023–24 Q5(c): which method for a real-time DAC with no non-causal filtering and no delay buffer? **Zero-order hold**: causal, needs only the current sample; compensate the sinc droop with an analog low-pass.

Worked example — CSE 311 2018–19 Q1(a)

Samples $g(0.01k) = \delta[k]$, g band-limited to 50 Hz ($f_s = 100$ Hz = Nyquist): $g(t) = \dfrac{\sin 100\pi t}{100\pi t}$, $G(f) = 0.01 \operatorname{rect}(f/100)$. As the band widens $g \to \delta(t)$ and $G \to 1$.

Traps

Time-limited signals are never exactly band-limited, so some aliasing is unavoidable. For a periodic signal $X[k] = Na_k$ holds only if $a_k = 0$ for $|k| \geq N/2$.

Part IV

Ashraf Sir — Laplace and z-Transforms

Chapter 9

The Laplace Transform

9.1 Definition, ROC and basic pairs

Key formulas

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,\mathrm{d}t = \mathcal{F}\{x(t)e^{-\sigma t}\}, \quad s = \sigma + j\omega; \qquad x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}\,\mathrm{d}s.$$

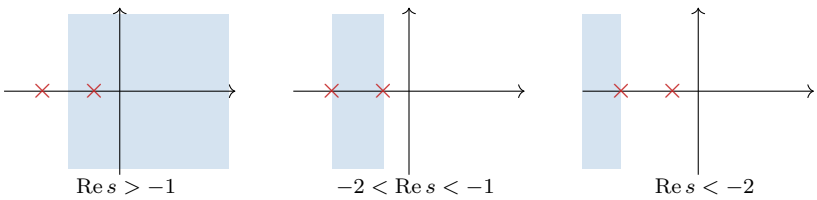
$s = j\omega$ gives the Fourier transform, valid only if the ROC contains the $j\omega$-axis. Eigenfunction: $e^{st} \to H(s)e^{st}$, $H = \mathcal{L}\{h\}$.
Unilateral (initial conditions at 0^-): $\mathcal{L}\{y'\} = sY - y(0^-)$, $\mathcal{L}\{y''\} = s^2Y - sy(0^-) - y'(0^-)$.

Pairs

$e^{-at}u(t)$	$\dfrac{1}{s+a}$	$\mathrm{Re}\, s > -a$
$-e^{-at}u(-t)$	$\dfrac{1}{s+a}$	$\mathrm{Re}\, s < -a$
$\delta(t)$	1	all s
$u(t)$	$\dfrac{1}{s}$	$\mathrm{Re}\, s > 0$
$t^{n-1}e^{-at}u(t)/(n-1)!$	$\dfrac{1}{(s+a)^n}$	$\mathrm{Re}\, s > -a$
$e^{-at}\cos\omega_0 t\, u(t)$	$\dfrac{s+a}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}\, s > -a$
$e^{-at}\sin\omega_0 t\, u(t)$	$\dfrac{\omega_0}{(s+a)^2+\omega_0^2}$	$\mathrm{Re}\, s > -a$
$e^{-b\lvert t\rvert}$, $b > 0$	$\dfrac{-2b}{s^2-b^2}$	$-b < \mathrm{Re}\, s < b$
$x = e^{-at}$ on $(0,T)$	$\dfrac{1-e^{-(s+a)T}}{s+a}$	all s (no pole at $-a$)

ROC rules

(1) Vertical strips. (2) No poles inside. (3) Finite-duration absolutely integrable: entire plane. (4) Right-sided: $\mathrm{Re}\, s > \sigma_0$. (5) Left-sided: $\mathrm{Re}\, s < \sigma_0$. (6) Two-sided: a strip (or empty). (7–8) Rational: bounded by poles; right-sided ⇒ right of the rightmost pole, left-sided ⇒ left of the leftmost pole.



Three ROCs of $X = \frac{1}{(s+1)(s+2)}$: three different signals.

Traps

The expression alone does not give the signal — always state the ROC. The ROC of a sum is the intersection of the ROCs (empty ⇒ no transform). A time shift does not change the ROC; pole cancellation can enlarge it.

LT and ROC of a composite signal

1. Split into right-sided, left-sided and finite pieces; transform each with its own ROC.
2. Use properties (shift e^{-st_0}, s-shift, $-tx \leftrightarrow X'(s)$, reversal $X(-s)$).
3. Intersect the ROCs; mark poles on the sketch.

Worked example — CSE 219 2023–24 Q6(b)

$x = |t|\,e^{-2|t|} - e^{-t}\cos 3t\,u(t) + t^2e^{-4t}u(t)$.
$te^{-2t}u \leftrightarrow \frac{1}{(s+2)^2}$ ($\operatorname{Re} s > -2$); $(-t)e^{2t}u(-t) \leftrightarrow \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{-1}{s-2}\right] = \frac{1}{(s-2)^2}$ ($\operatorname{Re} s < 2$); $e^{-t}\cos 3t\,u \leftrightarrow \frac{s+1}{(s+1)^2+9}$ ($\operatorname{Re} s > -1$); $t^2e^{-4t}u \leftrightarrow \frac{2}{(s+4)^3}$ ($\operatorname{Re} s > -4$).

$$X(s) = \frac{1}{(s+2)^2} + \frac{1}{(s-2)^2} - \frac{s+1}{(s+1)^2+9} + \frac{2}{(s+4)^3}, \qquad \boxed{-1 < \operatorname{Re} s < 2}.$$

Poles: -2 (double), 2 (double), $-1 \pm 3j$, -4 (triple). The ROC contains the $j\omega$-axis.

Worked example — 2022–23 Q1(b), Q2(c)

$x = e^{-at}$ on $0 < t < T$: $X = \frac{1-e^{-(s+a)T}}{s+a}$; at $s = -a$ the limit is T (finite), so $s = -a$ *is* in the ROC. $e^{-b|t-2|} \leftrightarrow \dfrac{2be^{-2s}}{b^2 - s^2}$, $-b < \operatorname{Re} s < b$.

9.2 Properties

Key formulas

Linearity	$ax_1 + bx_2$	$aX_1 + bX_2$	$\supseteq R_1 \cap R_2$
Time shift	$x(t - t_0)$	$e^{-st_0}X$	R
s-shift	$e^{s_0 t}x(t)$	$X(s - s_0)$	$R + \operatorname{Re} s_0$
Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X(s/a)$	aR
Reversal	$x(-t)$	$X(-s)$	$-R$
Convolution	$x_1 * x_2$	X_1X_2	$\supseteq R_1 \cap R_2$
Differentiation	$\mathrm{d}x/\,\mathrm{d}t$	sX	$\supseteq R$
s-differentiation	$-t\,x(t)$	$\mathrm{d}X/\,\mathrm{d}s$	R
Integration	$\int_{-\infty}^{t} x$	X/s	$\supseteq R \cap \{\operatorname{Re} s > 0\}$

Final value: if $sY(s)$ has all poles in the open LHP, $y(\infty) = \lim_{s\to 0} sY(s)$. Initial value: $x(0^+) = \lim_{s\to\infty} sX(s)$.

Differentiation

$x = \frac{1}{2\pi j}\int X e^{st}\,\mathrm{d}s \Rightarrow x' = \frac{1}{2\pi j}\int sXe^{st}\,\mathrm{d}s$, so $x' \leftrightarrow sX$ (e.g. $u \to \frac{1}{s}$, $\delta = u' \to 1$: the pole cancels and the ROC grows). s-domain: $\frac{\mathrm{d}X}{\mathrm{d}s} = \int(-t)x(t)e^{-st}\,\mathrm{d}t$. Final value: $\int_0^\infty y'e^{-st}\,\mathrm{d}t = sY - y(0^-) \to y(\infty) - y(0^-)$ as $s \to 0$.

9.3 Inverse Laplace transform

Partial fractions with the ROC

1. If $\deg N \geq \deg D$ divide first (polynomial part $\to \delta, \delta'$).
2. Factor D; expand: simple pole $\frac{A}{s+a}$ with $A = (s+a)X\big|_{s=-a}$; repeated poles $\frac{A_1}{s+a} + \frac{A_2}{(s+a)^2}$; complex poles $\to$ complete the square ($e^{-\alpha t}\cos\beta t$, $e^{-\alpha t}\sin\beta t$).
3. Use the ROC: a pole to the *left* of the ROC $\to$ right-sided term $Ae^{-at}u(t)$; a pole to the *right* $\to$ left-sided term $-Ae^{-at}u(-t)$.
4. Check with $x(0^+)$ or one numeric value.

Worked example — CSE 219 2022–23 Q2(b)

$X = \dfrac{s+2}{s^2+7s+12} = \dfrac{-1}{s+3} + \dfrac{2}{s+4}$, ROC $-4 < \mathrm{Re}\,s < -3$. Pole -3 is to the right of the ROC (left-sided), pole -4 to the left (right-sided):

$$x(t) = 2e^{-4t}u(t) + e^{-3t}u(-t).$$

Worked example — several inversions

$\frac{3s+7}{(s+1)(s+2)} = \frac{4}{s+1} - \frac{1}{s+2} \Rightarrow (4e^{-t} - e^{-2t})u$ $(\mathrm{Re}\,s > -1)$. $\frac{-5}{(s+1)(s-4)}$, $-1 < \mathrm{Re}\,s < 4$: $e^{-t}u(t) + e^{4t}u(-t)$. $\frac{s}{s+2} = 1 - \frac{2}{s+2} \Rightarrow \delta - 2e^{-2t}u$. $\frac{e^{-s}}{s(s+2)} \Rightarrow \frac{1}{2}[1 - e^{-2(t-1)}]u(t-1)$. $\frac{1}{(s-a)^3} \Rightarrow \frac{1}{2}t^2e^{at}u$.

9.4 LTI systems, differential equations, circuits, feedback

Key formulas

$Y = HX$. DE $\sum a_k y^{(k)} = \sum b_k x^{(k)}$: $H(s) = \dfrac{\sum b_k s^k}{\sum a_k s^k}$ (zeros from the numerator, poles from the denominator); the equation does *not* fix the ROC.
Causal $\Leftrightarrow$ ROC is a right half-plane. **Stable** $\Leftrightarrow$ ROC contains $j\omega$-axis; causal rational: all poles in the open LHP.
Cascade H_1H_2; parallel $H_1 + H_2$; negative feedback $\dfrac{G}{1+GH}$; disturbance after controller: $H_d = \dfrac{G}{1+G_cG}$.

Solving a DE with initial conditions (unilateral)

1. Transform: $Y\,D(s) = N(s)X(s) + I(s)$, with $I(s) = (s+a_1)y(0^-) + y'(0^-)$ for a second-order equation.
2. ZIR $= I/D$, ZSR $= NX/D$, $y =$ ZIR + ZSR.

3. Partial fractions, invert, check $y(0^+)$.

Worked example — CSE 219 2023–24 Q6(a)

$y''+7y'+12y=x'+2x$, $x=u$, zero ICs: $Y=\dfrac{s+2}{s(s+3)(s+4)}=\dfrac{1/6}{s}+\dfrac{1/3}{s+3}-\dfrac{1/2}{s+4}$,

$$y(t)=\left[\tfrac{1}{6}+\tfrac{1}{3}e^{-3t}-\tfrac{1}{2}e^{-4t}\right]u(t).$$

Worked example — EEE 211 2023–24 Q8(b)

$y''+5y'+6y=x'+x$, $y(0^-)=2$, $y'(0^-)=1$, $x=e^{-4t}u$. ZIR: $\frac{2s+11}{(s+2)(s+3)}\Rightarrow 7e^{-2t}-5e^{-3t}$. ZSR: $\frac{s+1}{(s+2)(s+3)(s+4)}\Rightarrow -\frac{1}{2}e^{-2t}+2e^{-3t}-\frac{3}{2}e^{-4t}$.

$$y=(6.5e^{-2t}-3e^{-3t}-1.5e^{-4t})u(t),\quad y(0^+)=2\ \checkmark.$$

Worked example — CSE 219 2023–24 Q6(c): ROC reasoning

Causal but unstable: $H=\frac{1}{s-2}$, ROC $\operatorname{Re}s>2$ (right half-plane $\Rightarrow$ causal; pole in RHP so the ROC excludes $j\omega$). Stable but non-causal: $H=\frac{-2}{s^2-1}$, ROC $-1<\operatorname{Re}s<1$, $h=e^{-|t|}$. For $H=\frac{s-1}{(s+1)(s-2)}$: $\operatorname{Re}s>2$ causal–unstable; $-1<\operatorname{Re}s<2$ stable–non-causal; $\operatorname{Re}s<-1$ anti-causal–unstable.

Circuits

1. Initial states: $v_C(0^-)$, $i_L(0^-)$ (for a source that is on at $t<0$, use the dc steady state at 0^-).
2. s-domain models: R; $L\to sL$ in series with $Li_L(0^-)$ (or parallel with $i_L(0^-)/s$); $C\to\frac{1}{sC}$ in series with $\frac{v_C(0^-)}{s}$.
3. KVL/KCL or division; partial fractions; invert; check $y(0^+)$.

Worked example — EEE 211 2023–24 Q4(a): series $L=2\,\mathrm{H}$, then $R=1\,\Omega\parallel C=2\,\mathrm{F}$, $y=v_C$

$v_C(0^-)=1$, $i_L(0^-)=2$, $x=u$: $\frac{1}{s}=2(sI-2)+V$, $I=(2s+1)V-2\Rightarrow V(4s^2+2s+1)=\frac{1}{s}+4s+4$.

$$Y=\frac{1}{s}+\frac{2}{4s^2+2s+1},\qquad y(t)=\left[1+\tfrac{2}{\sqrt{3}}e^{-t/4}\sin\tfrac{\sqrt{3}t}{4}\right]u(t).$$

Worked example — EEE 211 2023–24 Q5(b): steady-state with a controller

Unity feedback, plant $\frac{1}{s+2}$, $G_c=\frac{s+1}{s}$: $T=\frac{s+1}{s^2+3s+1}$, poles stable, $T(0)=1$ so $y(\infty)=A$ (integral action). With $G_c=\frac{1}{s-2}$: $T=\frac{1}{s^2-3}$ has a pole at $+\sqrt{3}$: unstable, the final-value theorem does not apply.

Worked example — other recurring checks

$y''+4y'+3y=u$, $y(0^-)=2$, $y'(0^-)=1$: $y=\frac{1}{3}+3e^{-t}-\frac{4}{3}e^{-3t}$. $y''+5y'+6y=2e^{-t}u$, $y(0^-)=3$, $y'(0^-)=-2$: $y=e^{-t}+5e^{-2t}-3e^{-3t}$. $H=\frac{s^2+5s+5}{s^2+5s+6}$: $h=\delta+(e^{-3t}-e^{-2t})u$. $H=\frac{2s+1}{(s+1)(s+2)}$: $h=(3e^{-2t}-e^{-t})u$. Closed loop with $H_1=\frac{1}{s^2+bs+b-3}$, $H_2=-2$: $H=\frac{1}{s^2+bs+b-5}$, stable for $b>5$.

In the exam

EEE papers: 2–3 Laplace questions every year (DE with ICs, RLC circuit, block diagram/feedback, final value). CSE 219: 2022–23 Q1–Q2 (32 marks), 2023–24 Q6 (35 marks). Figure-based questions: write $H(s)$ from the diagram first.

Chapter 10

The z-Transform

10.1 Definition, ROC and pairs

Key formulas

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \quad z = re^{j\omega}, \quad X(re^{j\omega}) = \mathcal{F}\{x[n] r^{-n}\}, \quad z = e^{sT}.$$

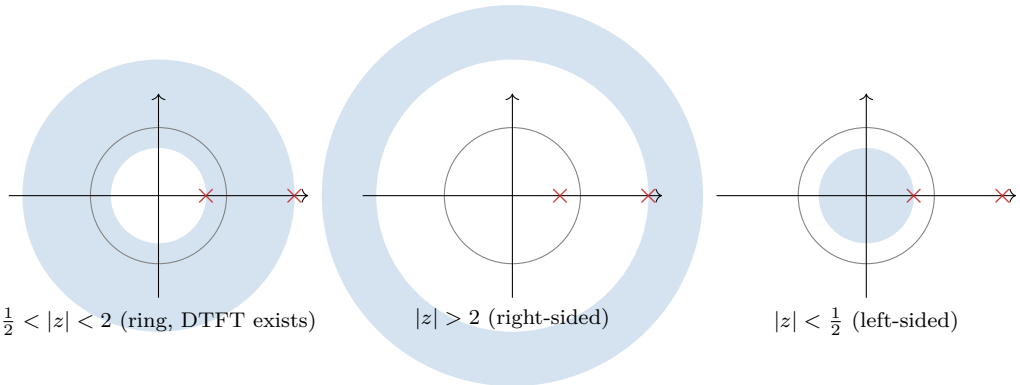
The DTFT exists iff the ROC contains $|z| = 1$. $y[n] = H(z) z^n$ for input z^n.

Pairs

$a^n u[n]$	$\dfrac{1}{1 - az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\dfrac{1}{1 - az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$\delta[n],\ \delta[n-1],\ \delta[n+1]$	$1,\ z^{-1},\ z$	all z; $z \neq 0$; finite z
$a^n,\ 0 \le n \le N-1$	$\dfrac{1 - a^N z^{-N}}{1 - az^{-1}}$	$z \neq 0$
$na^n u[n]$	$\dfrac{az^{-1}}{(1 - az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$b^{\lvert n\rvert},\ 0 < b < 1$	$\dfrac{1 - b^2}{(1 - bz^{-1})(1 - bz)}$	$b < \lvert z\rvert < 1/b$
$(\frac{1}{3})^n \sin(\frac{\pi}{4} n) u[n]$	$\dfrac{\frac{\sqrt{2}}{6} z^{-1}}{1 - \frac{\sqrt{2}}{3} z^{-1} + \frac{1}{9} z^{-2}}$	$\lvert z\rvert > \frac{1}{3}$

ROC rules

Ring centred at the origin; no poles inside; finite sequence: all z except 0 (terms with $n > 0$) and/or ∞ (terms with $n < 0$); right-sided $\Rightarrow$ exterior of a circle (includes ∞ iff causal); left-sided $\Rightarrow$ interior; two-sided $\Rightarrow$ annulus. Rational: right-sided outside the outermost pole, left-sided inside the innermost pole.



Poles at $\frac{1}{2}$ and 2; only the ring contains the unit circle (grey).

Worked example — CSE 219 2022–23 Q1(a), Q3(a)

$\sum_{n\geq 0} a^n z^{-n} = \frac{1}{1-az^{-1}}$ for $|az^{-1}| < 1 \Leftrightarrow |z| > |a|$. For $x[n] = a^n$, $0 \leq n \leq N-1$, $X(z)$ contains $az^{-1} \to \infty$ as $z \to 0$, so $0 \notin$ ROC; ROC is $|z| > 0$ (closed form $\frac{z(1-a^N z^{-N})}{z-a}$ has $N-1$ poles at the origin; the pole at a cancels).

10.2 Inverse z-transform

Key formulas

Contour: $x[n] = \dfrac{1}{2\pi j} \oint_C X(z) z^{n-1}\,\mathrm{d}z$ (C in the ROC, counter-clockwise). Derivation: inverse DTFT of $X(re^{j\omega})r^{-n}$, then $\mathrm{d}z = jz\,\mathrm{d}\omega$.

Inversion

1. Write X in powers of z^{-1} with unit constant terms; expand in partial fractions $\frac{A}{1-pz^{-1}}$.
2. $|p| <$ inner edge of the ROC (pole inside) $\to Ap^n u[n]$; pole outside the ROC $\to -Ap^n u[-n-1]$.
3. Irrational X (log): expand in a power series of z^{-1} valid in the ROC and read $x[n]$ as the coefficient of z^{-n}.
4. Causal check: $x[0] = \lim_{z\to\infty} X(z)$.

Worked example — CSE 219 2022–23 Q3(c): $X = \log(1+az^{-1})$, $|z| > |a|$

$\log(1+v) = \sum_{n\geq 1} (-1)^{n+1} \frac{v^n}{n}$ for $|v| < 1$; $v = az^{-1}$ satisfies this exactly for $|z| > |a|$:

$$x[n] = (-1)^{n+1} \frac{a^n}{n} u[n-1].$$

Worked example — CSE 219 2023–24 Q8(a)

$X = \dfrac{1+2z^{-1}-z^{-2}}{1-\frac{11}{6}z^{-1}+z^{-2}-\frac{1}{6}z^{-3}} = \dfrac{6}{1-z^{-1}} - \dfrac{3}{1-\frac{1}{2}z^{-1}} - \dfrac{2}{1-\frac{1}{3}z^{-1}}$, poles $1, \frac{1}{2}, \frac{1}{3}$.
$|z| > 1$: $x = [6 - 3(\frac{1}{2})^n - 2(\frac{1}{3})^n]u[n]$.
$\frac{1}{2} < |z| < 1$: $x = -6u[-n-1] - 3(\frac{1}{2})^n u[n] - 2(\frac{1}{3})^n u[n]$.
$\frac{1}{3} < |z| < \frac{1}{2}$: $x = -6u[-n-1] + 3(\frac{1}{2})^n u[-n-1] - 2(\frac{1}{3})^n u[n]$.
$|z| < \frac{1}{3}$: $x = -6u[-n-1] + 3(\frac{1}{2})^n u[-n-1] + 2(\frac{1}{3})^n u[-n-1]$.
The pole at $z = 1$ lies on the unit circle: no ROC contains it, so no DTFT exists.

10.3 Properties and LTI systems

Key formulas

Time shift	$x[n-n_0]$	$z^{-n_0}X$	R (except $0,\infty$)
z-scaling	$a_0^n x[n]$	$X(z/a_0)$	$\lvert a_0\rvert R$
Reversal	$x[-n]$	$X(z^{-1})$	R^{-1}
Expansion	$x[n/k]$	$X(z^k)$	$R^{1/k}$
Convolution	$x * h$	XH	$\supseteq R_1 \cap R_2$
First difference	$x[n]-x[n-1]$	$(1-z^{-1})X$	
Accumulation	$\sum_{k\le n} x[k]$	$\frac{X}{1-z^{-1}}$	$\supseteq R \cap \lvert z\rvert > 1$
z-differentiation	$nx[n]$	$-z\frac{\mathrm{d}X}{\mathrm{d}z}$	R
Initial value	$x[0]$ (causal)	$\lim_{z\to\infty} X$	

Causal $\Leftrightarrow$ ROC is an exterior including ∞. **Stable** $\Leftrightarrow$ ROC contains the unit circle. Causal rational: stable iff all poles inside $|z|=1$. Difference equation: $H = \dfrac{\sum b_k z^{-k}}{\sum a_k z^{-k}}$, ROC from causality.

Worked example — Difference equation

$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1]$: $H = \frac{1+\frac{1}{3}z^{-1}}{1-\frac{1}{2}z^{-1}}$, ROC $|z| > \frac{1}{2}$ (causal). Step response: $Y = \frac{8/3}{1-z^{-1}} - \frac{5/3}{1-\frac{1}{2}z^{-1}}$, $y[n] = \frac{8}{3}u[n] - \frac{5}{3}(\frac{1}{2})^n u[n]$.

Worked example — CSE 219 2023–24 Q8(b): pole–zero reasoning (poles $\frac{1}{3}, \frac{2}{3}, 2$)

statement	ROC
right-sided	$\lvert z\rvert > 2$
FT converges	$\frac{2}{3} < \lvert z\rvert < 2$
FT does not converge	$\lvert z\rvert > 2$, or $\frac{1}{3} < \lvert z\rvert < \frac{2}{3}$, or $\lvert z\rvert < \frac{1}{3}$
left-sided	$\lvert z\rvert < \frac{1}{3}$

Worked example — CSE 219 2023–24 Q8(c): $x[n] = |n|(\frac{1}{2})^{|n|}$

$n \ge 0$: $n(\frac{1}{2})^n u[n] \leftrightarrow \frac{\frac{1}{2}z^{-1}}{(1-\frac{1}{2}z^{-1})^2}$, $|z| > \frac{1}{2}$. $n < 0$: with $m = -n$, $\sum m(\frac{1}{2})^m z^m = \frac{\frac{1}{2}z}{(1-\frac{1}{2}z)^2}$, $|z| < 2$.

$$X(z) = \frac{2z(5z^2 - 8z + 5)}{(z-2)^2(2z-1)^2}, \quad \tfrac{1}{2} < |z| < 2.$$

Poles: $\frac{1}{2}$ (double), 2 (double); zeros: 0 and $0.8 \pm 0.6j$ (on the unit circle). Check $X(1) = 4 = \sum |n|(\frac{1}{2})^{|n|}$.

Worked example — CSE 219 2023–24 Q8(d): Laplace vs z

$\mathcal{L}$: continuous time, $s = \sigma + j\omega$, strips/half-planes, stability: ROC contains $j\omega$. $\mathcal{Z}$: discrete time, $z = re^{j\omega}$, rings, stability: ROC contains $|z| = 1$; $z = e^{sT}$. The map $s \to e^{sT}$ is many-to-one (s and $s + j2\pi/T$ give the same sequence), so z removes that redundancy and turns difference equations into algebra.

Appendix A

Question-Type Map and Exam Checklist

A.1 Which method for which question (recent papers)

Question type	Go to	Recent TF
midrule Sketch $x(\alpha t+\beta)$, sums of shifted/scaled pieces	Ch. 1 §1.2 recipe, $t=(a-\beta)/\alpha$	22-23 Q7; 23-24 Q1(a)
Even–odd decomposition + graphs	Ch. 1 §1.3	22-23 Q5(a)
Periodicity (DT/CT)	Ch. 1 §1.5	22-23 Q5(c); 23-24 Q3(a)
System properties, interconnections, h_{eq}	Ch. 2 §2.1–2.2	23-24 Q1(b)
DT/CT convolution, interval table	Ch. 2 §2.3	22-23 Q8(a)
Causal? stable?	Ch. 2 §2.4	22-23 Q8(b); 23-24 Q3(b)
FS coefficients, shift/derivative properties	Ch. 3	23-24 Q4
FT of a signal with properties, spectra	Ch. 4	22-23 Q5(b), Q6; 23-24 Q1(c), Q2
8-pt DIT FFT, matrix FFT, Bluestein	Ch. 6	23-24 Q5(a), Q7(a),(b)
Linear convolution via DFT	Ch. 5	23-24 Q5(b)
Sampling theorem, aliasing, reconstruction	Ch. 7	22-23 Q1(c), Q4(b); 23-24 Q5(c), Q7(c)
LT + ROC, inverse LT, properties	Ch. 8	22-23 Q1(b), Q2; 23-24 Q6
z-transform, ROC, inverse	Ch. 9	22-23 Q1(a), Q3; 23-24 Q8

A.2 Universal exam checklist

Before you write

1. Read the whole question; underline what is asked (draw, prove, state the ROC, name the property).
2. Sketch first — signals, overlap, spectra, ROC strips and pole–zero plots.
3. Write steps in your own compact way; show intermediate signals/matrices when the question says "showing all stages".
4. Check at the end: units/limits, continuity at interval boundaries, $X[0]=\sum x$, $y(0^+)$, support, stability of poles.

5. Always state the ROC next to every Laplace and z answer.

Frequent mistakes

- Order of shift and scaling; forgetting the reversal when $\alpha < 0$.
- Dropping the $n < 0$ (or $t < 0$) region of a convolution.
- Impulses at jumps lost when differentiating.
- a_0 missing after integrating or dividing by $jk\omega_0$.
- Wrong twiddle: $W_8^3 = -\frac{1+j}{\sqrt{2}}$, $W_4 = -j$ (not $+j$).
- Giving a Laplace/z expression without the ROC; picking the wrong-sided term for a pole.
- Using the final-value theorem when poles lie in the right half-plane or on the $j\omega$-axis.